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Several Measures of Trophic Structure Applicable to Complex Food Webs

STEPHEN LEVINE

Department of Electrical Engineering, Merrimack College, North Andover, MA 01845, *U.S.A.*†

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A continuous measure of "trophic position" is introduced, based on average function of an ecosystem component. Two measures of trophic position variance are defined, and the ideas of trophic specialists and generalists are introduced. The analysis is based on a Markov chain model of energy flows. Along with a number of simple ecosystem structures, the model is also applied to data on the North Sea ecosystem. The model developed in this paper allows for describing in trophic level terms ecosystems which differ substantially from food chains. It is envisioned that such a description would play a useful role in the comparative analysis of ecosystems.

1. Introduction

Since being introduced by Lindeman (1942) the concept of trophic levels has been a central theme in ecology. Trophic levels partition an ecosystem into stages of energy processing (Odum, 1971, p. 61). A trophic level is therefore not a collection of species; rather, it is an ecosystem function. Individual species can, and often do, perform functions corresponding to two or more trophic levels (Odum, 1975, p. 64; Ricklefs, 1973, p. 643). Only in simple ecosystem structures, most notably the food chain, is there a simple correspondence between trophic levels and species.

Most real ecosystem structures are complex food webs, thus they are not readily described in trophic level terms. Schemes to overcome this by emphasizing one trophic function over the others have been suggested (Roberts, 1976; Harary, 1961). As noted by Ulanowicz & Kemp (1978), this involves a tradeoff between the need to consolidate in order to present a useful system description and the need to minimize functional ambiguity.

A number of investigators have suggested another approach, that of assigning the attributes of a species, or more correctly a system component, to different trophic levels in amounts proportional to its different functions (Cummins *et al.*, 1966; Kemp & Homer, 1978; Ulanowicz & Kemp, 1978). Proportionality is measured by the fraction of input energy it receives over

† Present address: Department of Engineering Design, Tufts University, Medford, MA 02155, U.S.A.

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paths of different lengths. Kemp & Homer (1978) present a mathematical technique for significantly simplifying system structure so that the number of components is determined by the number of trophic levels. Ulanowicz & Kemp (1978) pursue this further in order to reduce the system to a food chain, thus allowing unambiguous trophic levels to be defined. This last goal is achieved at some cost in terms of biological meaningfulness.

A different goal is pursued in this paper, again based on the relative importance of a component's various trophic functions. Rather than redefining system components, the existing components are characterized by measures of their role within the trophic structure. Trophic position, the first of these measures, reflects "average" function of a component. Additional measures are introduced to provide further description.

A related measure of trophic function was developed by Kercher & Shugart (1975). Termed "effective trophic position", it was defined as "a function of the net energy input to the food web necessary to deliver one unit of energy flux" to the ecosystem component being considered. Thus energy transfer efficiency is included in this definition.

The model developed here is based, as are all trophic-dynamic models, on energy flow in ecosystems. The dynamics of energy flow are very different than the dynamics of populations. Community energetics may remain stable even when species vary greatly in population density (McNaughton, 1977). Thus competition between two species, though a critical question from the evolutionary viewpoint, may have minimal impact on localized considerations of energy flow. It is, however, unlikely that a change in energy flow will not result in substantial changes in population structure; this is of great importance in ecosystem management and environmental impact analysis.

While numerous ecosystem models based on energy flow have been developed, that of Hannon (1973) is of particular interest here. This model utilizes a modified input–output analysis (Yan, 1969). Building on this model Finn (1976*a*,*b*) and Patten & Finn (1977) have described several system parameters of energy and nutrient flow. Ulanowicz & Kemp (1978) also use this formulation as their starting point, as well as in reformulating related work of Kemp & Homer (1978).

An alternative to the input–output formulation, and the approach utilized in this paper, is provided by the theory of Markov chains (Roberts, 1976, p. 320; Kemeny & Snell, 1960). Finn (1976*b*) and Barber (1978*a*,*b*) have developed Markovian models with different goals than pursued here. Their interest centers on describing flows; in this paper flows are used to classify system components.

The term component, or alternatively compartment, needs further explanation. Components may be biological energy processors such as species, or energy sources, biological or not, such as the sun or nutrients flushed into a lake. Biological components include species, groups of species, or subsets of a species, such as juveniles or demes. Choice of components can strongly influence the system structure and is based, at least in part, on the goals of the analysis.

The plan of this paper is as follows. Following basic considerations of trophic structure an energy budget model is introduced. This is easily converted to a transition matrix model allowing interpretation as a Markov chain, in particular one containing absorbing states. The theory of absorbing chains leads to results interpretable as trophic position measures. The concept of trophic specialization is introduced, and several simple, and hopefully illustrative, examples are considered. Finally, the technique is applied to an example of actual data.

2. Considerations of Trophic Structure

The simplest trophic structure, and in fact the paradigm, is the food chain. The digraph in Fig. 1 shows an example, with S_1, S_2 and S_3 the components. S_1 belongs to trophic level 0, S_2 to level 1 and S_3 to level 2; the arrows correspond to energy flows. Ultimately any food web structure can be reduced to this form by appropriate definition of the components, and in fact Fig. 1 defines what is meant by trophic levels.

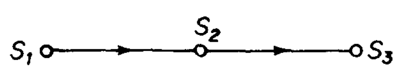

FIG. 1.

More complex and detailed food webs may still allow for unambiguous associations of components with trophic levels, requiring only that all paths between any two components be of equal length, that is, have the same number of branches. Figure 2 indicates such an example; component S_1 is

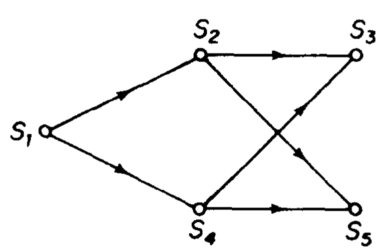

FIG. 2.

trophic level 0; S_2 and S_4 constitute trophic level 1; S_3 and S_5 constitute trophic level 2.

When all paths between two components are not of equal length the food chain model is not strictly followed, and components cannot be unambiguously assigned to trophic levels. In Fig. 3 component S_3 functions at both trophic levels 1 and 2. If we are to more accurately assess this component's position in the trophic structures we require additional information regarding the relative importance of the two roles; specifically we will wish to assign weights to the digraph branches so that an "average" function can be determined.

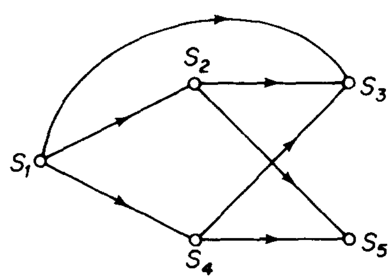

FIG. 3.

An even more complex situation occurs if the food web includes cycles, as in Fig. 4. Paths of great, in fact infinite, length exist. Basic considerations of energy use efficiency and conservation indicate that these longer paths must be of relatively less weight. Nonetheless, these cycles do contribute to the determination of trophic position.

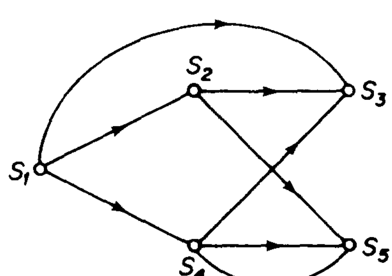

FIG. 4.

Since trophic function depends on the source of a component's energy an energy budget model provides an appropriate way of weighting the various branches. Let g_{ij} be that fraction of its energy input which component j

derives from component i. This essentially defines the **G** matrix of Hannon (1973, p. 537). Figure 5 represents an example. Two component types, with corresponding nodes, are evident; source components, in this case node 1, with no inputs; and energy processing components, all other nodes, for which $\sum_i g_{ij} = 1$. All energy is ultimately due to the source nodes.

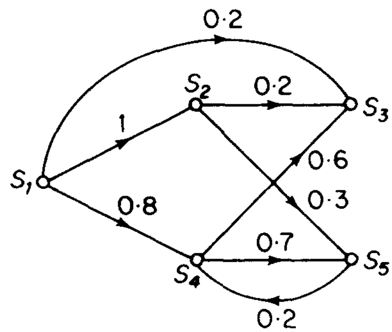

FIG. 5.

An energy budget model, such as in Fig. 5, can also be interpreted as an absorbing Markov chain (Kemeny & Snell, 1976, Ch. 3; Roberts, 1976), the branch values then representing probabilities. If we assume the source node receives all its energy from itself, Fig. 5 can be described by the transition matrix **T**,

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0{\cdot}2 & 0{\cdot}2 & 0 & 0{\cdot}6 & 0 \\ 0{\cdot}8 & 0 & 0 & 0 & 0{\cdot}2 \\ 0 & 0{\cdot}3 & 0 & 0{\cdot}7 & 0 \end{bmatrix}$$

where t_{mn} is the probability that a unit of energy entering component m was obtained directly from component n. Thus t_{41} indicates the 80% probability that energy taken in by component 4 came directly from the source. Node 1 represents the source, or in Markov chain terminology, the absorbing state. Once energy has been traced back to a source, it is assumed to have been there for all previous time.

3. Trophic Position Measures

Trophic position is intended to measure average trophic function, that is, the expected, or mean, length of the path over which a component obtains energy from the source. Remember that the probability that energy is obtained over a given path is the product of the branch probabilities contained in that path. Since all $t_{mn} \leq 1$, longer paths generally have smaller probabilities associated with them.

Specifically, let K be the random variable representing a path length from the source to a given component, and $p_i(k)$ be its probability function for the ith component. Note, $p_i(k)$ can be the sum of one or more paths of length k. In Fig. 5, $p_3(0)=0$, $p_3(1)=0{\cdot}2$, $p_3(2)=(0{\cdot}2\times 1)+(0{\cdot}6\times 0{\cdot}8)=0{\cdot}68$, $p_3(3)=0$, $p_3(4)=(0{\cdot}6\times 0{\cdot}2\times 0{\cdot}7\times 0{\cdot}8)+(0{\cdot}6\times 0{\cdot}2\times 0{\cdot}3\times 0{\cdot}1)=0{\cdot}1032$, etc. The trophic position of component i, x_i, is then the expected value of K;

$$x_i = \sum_{k=0}^{\infty} k p_i(k). \tag{1}$$

Noting that $p_3(0)$ to $p_3(4)$ includes 98% of the energy, we can approximate x_3 as $x_3 = (0\times 0) + (1\times 0{\cdot}2) + (2\times 0{\cdot}68) + (3\times 0) + (4\times 0{\cdot}1032) = 1{\cdot}97$. Inclusion of longer paths gives an answer of approximately $x_3 = 2{\cdot}1$.

An alternative, and computationally more efficient, method exists for computing the trophic position vector $\mathbf{x}$. It is equivalent to the procedure for computing the expected number of transitions before absorption in an absorbing chain (Kemeny & Snell, 1960, p. 43–51). However, it will be presented here in ecological terms.

If necessary, the system components are renumbered so that the source components come first. Then the $\mathbf{T}$ matrix can be partitioned as

$$\mathbf{T} = \begin{bmatrix} I & 0 \\ R & Q \end{bmatrix} \tag{2}$$

where the $\mathbf{Q}$ matrix pertains to flows between non-energy sources and the $\mathbf{R}$ matrix pertains to flows from the sources to the non-energy source components. The vector $\mathbf{x}$ can also be partitioned. By definition the source components are trophic position zero; therefore

$$\mathbf{x} = \begin{bmatrix} \mathbf{0} \\ \mathbf{y} \end{bmatrix} \tag{3}$$

where $\mathbf{y}$ is the trophic position vector for the non-energy source components.

We would expect the trophic position of a component to be one greater than the weighted average of the trophic positions of its resources. Certainly, this is consistent with trophic level concepts. Putting this in mathematical terms,

$$\mathbf{y} = R\mathbf{0} + Q\mathbf{y} + \mathbf{1} \tag{4}$$

and solving for $\mathbf{y}$;

$$\begin{aligned}\mathbf{y} &= (I-Q)^{-1}\mathbf{1} \\ &= N\mathbf{1}.\end{aligned} \tag{5}$$

Equation (5) states that the trophic positions can be calculated from the row sums of $N=(I-Q)^{-1}$ (see theorem 3.3.5, p. 51 in Kemeny and Snell, 1960). As an example, for $x_3 = y_2$ we would obtain 2·08. The results from equation (5) are consistent with those of equation (1).

4. Trophic Specialization

Trophic position measures average function of a component based on energy flows. This average may be achieved by obtaining energy over paths of similar or dissimilar length. The variance of path length K can be utilized as a measure of trophic specialization; thus we may have trophic specialists and generalists. The suitable measure is

$$\sigma_i^2 = \sum_{k=0}^{\infty} (k-x_i)^2 p_i(k). \tag{6}$$

If we apply this measure to a food chain or any other food web for which unique trophic levels can be assigned, we will find it to be zero for all components, as expected. A rather more interesting aspect of this definition exists; a resource specialist, that is, a component utilizing only one other component, need not be a trophic specialist. This occurs when that resource component is itself not a trophic specialist.

Alternatively, a trophic specialist need not be a resource specialist. Herbivores are essentially trophic specialist, though many consume a wide variety of plants. Plants themselves are trophic specialist, thus so are their consumers.

A second model of trophic specialization can be proposed, related to equation (5). Rather than considering path lengths, it considers the resources utilized by a component. If all resources used have the same trophic position, the component is a trophic specialist. Specifically, this measure is based on the variance of the trophic positions of the utilized resources, where each resource is weighted by its level of utilization. The weighted average of resource trophic positions utilized by component i is, consistent with the derivation of equation (5),

$$r_i = x_i - 1 \tag{7}$$

and the variance determining trophic specialization is

$$\hat{\sigma}_i^2 = \sum_j (x_j - r_i)^2 t_{ij}. \tag{8}$$

This variance is zero for both resource specialist and trophic specialist as defined by equation (6). Equations (6) and (8) give related, but different information on the trophic structure of an ecosystem.

5. Several Simple Numerical Examples

This section contains several numerical examples, utilizing the trophic structures discussed in section 2. Figure 2 is repeated in Fig. 6.

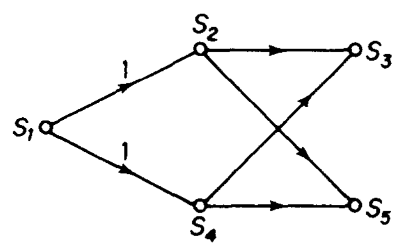

FIG. 6.

The trophic position vector, $\mathbf{x}$, and standard deviations, $\boldsymbol{\sigma}$ and $\hat{\boldsymbol{\sigma}}$, are $\mathbf{x} = (0\ \ 1\ \ 2\ \ 1\ \ 2)'$, $\boldsymbol{\sigma} = \hat{\boldsymbol{\sigma}} = (0\ \ 0\ \ 0\ \ 0\ \ 0)'$. These values reflect the unambiguous nature of trophic function for this case. Components 2 and 4 are associated with trophic level 1 and components 3 and 5 with trophic level 2.

Next, consider the effect of direct energy flow from source component 1 to component 3, as shown in Fig. 7.

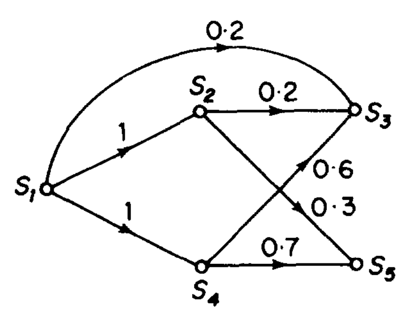

FIG. 7.

This leads to $\mathbf{x} = (0 \;\; 1 \;\; 1{\cdot}8 \;\; 1 \;\; 2)'$ and $\boldsymbol{\sigma} = \hat{\boldsymbol{\sigma}} = (0 \;\; 0 \;\; 0{\cdot}4 \;\; 0 \;\; 0)'$, once again the two standard deviations being equal. The direct link lowers the trophic position of component 3, at the same time resulting in its having a non-zero standard deviation.

An example with energy cycling is shown in Fig. 8. In this case $\mathbf{x} = (0 \;\; 1 \;\; 2{\cdot}28 \;\; 1{\cdot}47 \;\; 2{\cdot}33)'$, $\boldsymbol{\sigma} = (1 \;\; 1 \;\; 0{\cdot}80 \;\; 0{\cdot}99 \;\; 0{\cdot}85)'$ and $\hat{\boldsymbol{\sigma}} = (0 \;\; 0 \;\; 0{\cdot}22 \;\; 0{\cdot}93 \;\; 0{\cdot}13)'$. The cycling causes an increase in trophic position, compared to Fig. 6, for all components whose nodes are in, or reachable from, the cycle. This increase in trophic positions represents the energy cycled through many stages of energy processing. All components affected by the cycle have non-zero standard deviations as well.

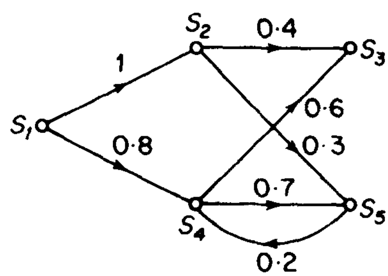

FIG. 8.

A comparison of $\boldsymbol{\sigma}$ and $\hat{\boldsymbol{\sigma}}$ is interesting. In particular note that $\hat{\sigma}_3$ and $\hat{\sigma}_5$ are much smaller than σ_3 and σ_5 while $\hat{\sigma}_4$ and σ_4 are similar. This reflects the essential difference between the two standard deviation measures. σ_3 and σ_5, as well as σ_4, indicate that components 3, 4 and 5 utilize energy obtained from paths of greatly varying lengths. $\hat{\sigma}_3$ and $\hat{\sigma}_5$, in turn, indicate that these respective components directly obtain their energy from components of similar trophic position. Component 4 receives its energy from components of dissimilar trophic position, thus $\hat{\sigma}_4$ is larger. Thus, the two measures, $\boldsymbol{\sigma}$ and $\hat{\boldsymbol{\sigma}}$, provide complementary information about the structure of an ecosystem.

The final example considered, shown in Fig. 9, combines the direct flow to component 3 and cycling between components 4 and 5. As a result, $\mathbf{x} = (0 \;\; 1 \;\; 2{\cdot}08 \;\; 1{\cdot}47 \;\; 2{\cdot}33)'$ $\boldsymbol{\sigma} = (0 \;\; 0 \;\; 0{\cdot}95 \;\; 0{\cdot}99 \;\; 0{\cdot}85)'$ and $\hat{\boldsymbol{\sigma}} = (0 \;\; 0 \;\; 0{\cdot}57 \;\; 0{\cdot}93 \;\; 0{\cdot}13)'$. Note that x_3 is reduced from the previous example; there is a cancelling of trophic position shift due to the direct flow and cycling. However, both σ_3 and $\hat{\sigma}_3$ are larger than ever; the contributions to standard deviation of direct flow and cycling reinforce each other. Component 3 is, on the average, functioning at near the second trophic level.

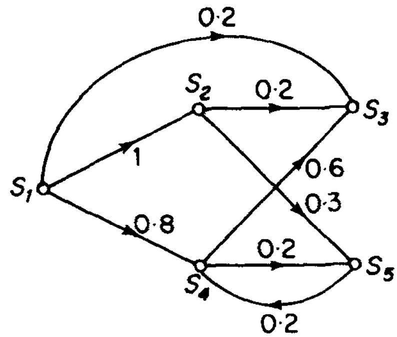

FIG. 9.

In fact, 68% of its energy input is in the second stage of processing, 20% in the first stage, and the remaining 12% in the third or greater stage.

6. North Sea Food Web

Steele (1974) presented data for many energy flows in the North Sea food web. Ulanowicz & Kemp (1978) subsequently added the remaining flows by *ad hoc* guesses, and applied their aggregation procedure to the system. This data can be interpreted in terms of the model developed here, noting that the components in Steele's food web are already species aggregates rather than

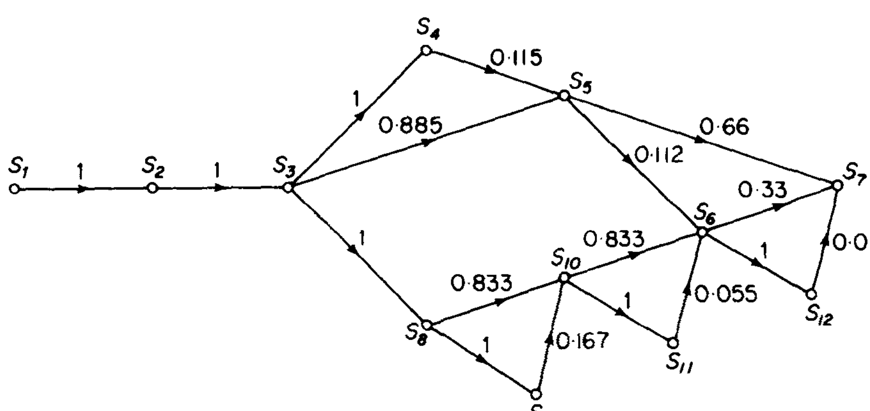

FIG. 10. North Sea food web. S_1, energy source; S_2, primary producers; S_3, pelagic herbivores; S_4, invertebrate carnivores; S_5, pelagic fish; S_6, dimersal fish; S_7, man; S_8, bacteria; S_9, meio-benthic; S_{10}, macro-benthic; S_{11}, other carnivores; S_{12}, large fish.

single species. If the feeding behavior of all species within a component is similar, then the trophic position of the component is a good approximation for all the included species.

Steele's data included energy flows to man from the food web. This is included in our analysis, recognizing that the resultant trophic position for man refers only to the food web considered.

Figure 10 presents the appropriate flow graph.

While the North Sea model includes no cycles the web differs from a food chain. Table 1 indicates that man, in spite of being the "top predator" does

TABLE 1

Trophic position and standard deviations for North Sea food web

	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	S_9	S_{10}	S_{11}	S_{12}
$\mathbf{x}$	0	1	2	3	3·12	5·11	4·8	3	4	4·17	5·17	6·11
σ	0	0	0	0	0·32	0·55	1·04	0	0	0·37	0·37	0·55
$\hat{\sigma}$	0	0	0	0	0·32	0·42	0·96	0	0	0·37	0	0

not have the highest trophic position. This results from his utilization of many relatively short energy paths, as well as some long ones. Not surprisingly, by either measure man has the greatest trophic variance; in fact, he utilizes three components directly, and is the final component in fourteen "chains" ranging in length from four to nine branches, the shortest being the most important. The North Sea model also indicates the degree to which components of similar trophic position can differ in trophic specialization. Thus, man and dimersal fish differ only 6% in trophic position, but man has twice the standard deviation, reflecting a very different predatory role. This is true in spite of the fact that each utilizes three other components. Man utilizes them more equally, and he utilizes components of different trophic position.

7. Further Discussion

The study of ecosystem structure, whether in terms of energy flow or not, addresses a number of problems. First is the role of different species, or more generally components, within the system itself. Second is the comparison of different ecosystems; which species play similar roles, or the possibility that the same species may play different roles. The recent interest in ecosystem convergence (Peet, 1978; Mooney, 1977; Cody, 1974) is an example of these concerns. Another area is the transition of ecosystem structure, whether it is short term good seasonal, long term successional or human induced, phenomena. Whatever the nature of the study, the ability to quantify descriptions is one of great potential value.

It is not only individual components that can be compared; measures pertaining to the complete structure are possible. Average trophic position, and the distribution of trophic position are indicators of overall structure. The variance measures indicate the level of complexity of the ecosystem; the average variance is a possible measure of this type. For example the structures in Figs 6–9 have σ_{ave} equal to 0, 0·08, 0·53, and 0·56 respectively, while the $\hat{\sigma}_{ave}$ are 0, 0·08, 0·26, and 0·33. The average standard deviations reflect the growing complexity. By comparison the North Sea ecosystem is characterized by $\sigma_{ave} = 0{\cdot}27$ and $\hat{\sigma}_{ave} = 0{\cdot}17$; the absence of cycles leads to these relatively small values. On the other hand x_{ave} for the North Sea ecosystem is 3·46 as compared to 1·42 for Fig. 7, the highest of the simple examples. In many ways the North Sea ecosystem is not far from being a food chain. This may, in turn, simply reflect the broad definitions of system components used.

The biological significance of trophic specialization may be one of great interest. The following example is presented merely as an illustration.

Figure 11 presents two contrasting interaction models (see Levine, 1977 for details on this type model). In both diagrams the top predator, P_A or P_B, is a generalist. Fig. 11(a) indicates that P_A is a resource generalist but trophic specialist. The detrimental effect P_A has on S_1, S_2 or S_3 due to predation is balanced by the beneficial effect of reducing competition. This is analogous to the situation discussed by Paine (1974) where predation encourages community diversity.

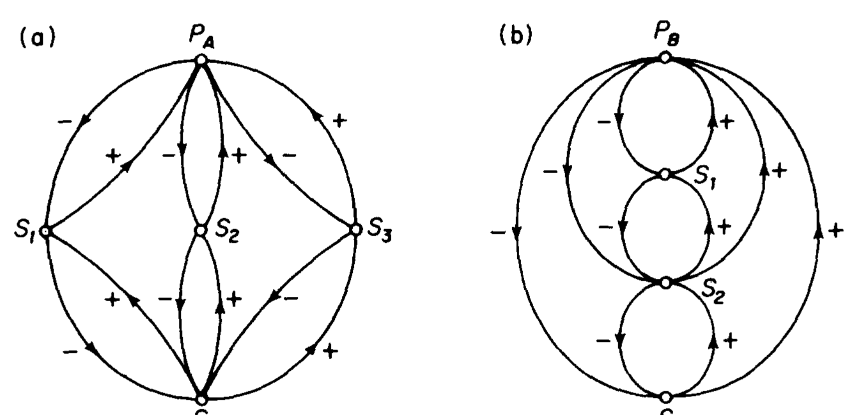

FIG. 11.

In Fig. 11(b) a different situation exists. P_B is both a resource and trophic generalist. Again balancing interactions occur, but note that P_B is both a predator and a competitor of species S_1, both detrimental to S_1. The same holds for the effect of P_B on S_2, except that this is somewhat balanced by the reduced predation on S_2 by S_1. In general trophic generalist do not have the same symmetrical and moderating interactions with their resources as exists for trophic specialists. Their effect on community diversity is somewhat less clear.

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